Predicting the carbon emissions of a European multi-market automotive portfolio

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Abstract

Measuring the carbon footprint of asset portfolios has become both a regulatory (CSRD, EU green taxonomy) and a strategic requirement for financial institutions. For a vehicle-financing portfolio, this footprint depends first and foremost on the carbon intensity (gCO_2/km) of the financed vehicles, a homologation quantity that is frequently missing from management databases. We tackle its prediction on a real, heterogeneous, multi-country dataset (seven European markets, $\approx$ 800k observations) that is far removed from the clean, single-source homologation datasets typically used in the literature. We propose a complete, auditable machine-learning chain: (i) fuzzy-matching imputation of missing technical features, whose distributional impact is controlled via Jensen-Shannon divergence and Wasserstein distance; (ii) a four-model benchmark (LightGBM in two feature scopes, XGBoost, an MLP) under 5-fold cross-validation, where SHAP analysis serves as a diagnostic tool and motivates a feature-scope reduction that curbs overfitting; (iii) distribution-free conformal prediction intervals (MAPIE), directly usable for bounded reporting; and (iv) a cross-market transferability study via a domain classifier, a PCA-based cluster analysis and covariate-shift AUCs, establishing strong covariate shift but no concept shift, hence correctable by importance reweighting. Throughout, explainability and auditability are treated as first-class constraints, on par with predictive accuracy.

1 Context and methodological framework

Motivation. Measuring the carbon footprint of asset portfolios has become both a regulatory (CSRD, EU green taxonomy) and a strategic requirement for financial institutions. For a vehicle-financing portfolio, the objective is twofold: to produce a reliable and *bounded* estimate of the aggregate footprint for reporting purposes, and to provide a reproducible, explainable and transferable methodology, one that internal teams can audit and take over. A model that is accurate but opaque, that does not generalise, or that cannot be applied to real data would have no operational value. It is the tension between statistical rigour and usability that structures our choices: explainability (SHAP, §2.2.1) and uncertainty quantification (conformal prediction, §2.3.4) are treated as first-order constraints. Over a portfolio of several hundred thousand vehicles, even a small systematic error on intensity propagates into a large aggregate bias on the reported footprint; accuracy *and* control of the associated uncertainty are therefore inseparable.

Data and decoupling. The portfolio covers seven European markets, with completeness rates and encoding conventions that vary from one market to the next. The annual footprint of a vehicle factorises as

$$E_{CO_2} = D_{\text{annual}} \times I_{\text{carbon}}, \tag{1}$$

where I_{carbon} (gCO_2/km) depends on the physical and technical characteristics and is measured at homologation, whereas D_{annual} depends on usage. These determinants are disjoint, which allows the two to be predicted separately. Since intensity is partially present in the data (distance, by contrast, is not), we focus this work on I_{carbon}; predicting D_{annual} is an extension beyond the present scope.

State of the art. The prediction of CO_2 emissions by ML methods has been the subject of a growing but heterogeneous literature since 2020. Two families of problems coexist: prediction from the static technical characteristics of the vehicle, and prediction of instantaneous emissions from dynamic driving data (speed, acceleration, road gradient) [20]. The conclusions diverge: gradient boosting (XGBoost, LightGBM, CatBoost) generally dominates on static tabular data [13, 5], but some works favour dedicated neural networks [1], and many comparisons are biased by circular features (typically the measured fuel consumption, used as a predictor of emissions even though it is unavailable in a banking context). More fundamentally, the datasets used are almost exclusively public homologation databases (Canada, United Kingdom, European Union), clean, complete and single-source, with no relation to a multi-country banking portfolio that is heterogeneous, has a high rate of missing values and variable encoding conventions. Dataset shift between markets, although critical as soon as an international model is contemplated, is addressed in none of the works identified; as for the annual distance, the few existing approaches rely on on-board telematics data [11] or mobility surveys, neither of which is accessible in our setting. It is precisely in these blind spots, robustness to degraded data, multi-market generalisation and uncertainty quantification, that our contributions lie.

2 Predicting carbon intensity

2.1 Pre-processing and imputation

Harmonisation and outliers. The data come from heterogeneous national databases, each with its own conventions for categorical labels (`FUEL_TYPE` in particular) and continuous ones (`HORSEPOWER`, expressed in mechanical or fiscal horsepower depending on the country), which must first be normalised to a common reference. Outliers are then removed ($CO_2 > 500$ gCO_2/km, incompatible with the WLTP/NEDC standards, or `ENGINE_SIZE` manifestly inconsistent with the other characteristics).

Imputation of missing features. The fields `ENGINE_SIZE` (displacement) and `ENGINE_WEIGHT` (kerb weight) have a non-negligible rate of missing values, particularly for older vehicles and in certain countries. They are among the most predictive features (see SHAP, §2.2.1): their absence would severely degrade the coverage of the model. Yet these quantities are quasi-deterministic within a (`BRAND`, `MODEL`) cluster, since a given production model is offered with only a small number of powertrains. The within-group variance is small relative to the between-group variance (Fig. 1): the mean conditional on (`BRAND`, `MODEL`) is therefore a low-bias estimator. This assumption is validated by cross-validation: a fraction of the observed values is masked and imputed, then the imputed values are compared with the held-out ones, which serve as a reference (this complements the distributional metrics below). Since the within-cluster variance is also known, the imputation could have been refined by sampling from a normal distribution with the same mean and variance as the cluster, rather than using the conditional mean alone. The pipeline operates in two stages: (i) fuzzy matching, where a dictionary mapping each key (`BRAND`, `MODEL`) to the observed mean is built, and imputation is carried out through the closest key in the sense of the Levenshtein distance (above a similarity threshold); (ii) a fallback on the mean conditional on `BRAND` or `FUEL_TYPE` when no sufficient match is found.

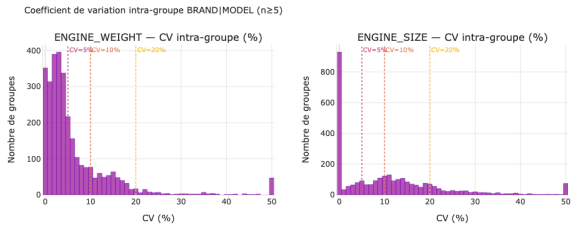


Figure 1. Validation of the imputation: within-cluster dispersion of `ENGINE_SIZE` and `ENGINE_WEIGHT` inside the (`BRAND`, `MODEL`) groups. The low within-cluster variance, compared with the between-group variance, justifies imputation by the conditional mean.

Distributional control. To check that the imputation does not distort the distributions, an auditability requirement, we compare the before (P) and after (Q) imputation states through two complementary metrics. The Jensen-Shannon divergence, a symmetric extension of Kullback-Leibler,

$$\mathrm{JS}(P\|Q) = \tfrac{1}{2}\mathrm{KL}(P\|M) + \tfrac{1}{2}\mathrm{KL}(Q\|M), \quad M = \tfrac{P+Q}{2}, \quad (2)$$

measures dissimilarity (close to 0: no distortion). The Wasserstein distance, $W_1(P,Q) = \inf_{\gamma\in\Gamma(P,Q)} \mathbb{E}_{(x,y)\sim\gamma}|x-y|$, being sensitive to the geometry of the space, penalises mass displacements in proportion to their amplitude and is therefore informative about the *magnitude* of the biases. We take $\mathrm{JS} < 0.05$ and W_1 below 5 % of the interquartile range as acceptability thresholds. The results (Table 1) are satisfactory: with the exception of `ENGINE_SIZE` in Spain (5.44 % imputed, $W_1 = 15.12$), the distortions remain marginal.

Table 1. Rate of imputed missing values by market, and distributional control metrics (`WGT` = `ENGINE_WEIGHT`, `SIZE` = `ENGINE_SIZE`).

Country	Obs.	WGT	JS	W_1	SIZE	JS	W_1
Ger.	48 246	1.94%	9×10^{-5}	4.98	0.38%	2×10^{-4}	2.15
Fra.	199 823	2.43%	2×10^{-5}	2.30	2.42%	3×10^{-4}	3.94
Ita.	236 643	1.24%	5×10^{-5}	2.30	1.24%	3×10^{-5}	2.86
Por.	–	–	–	–	–	–	–
Spa.	308 381	2.38%	5×10^{-4}	3.89	5.44%	4×10^{-3}	15.12

Exploratory analysis. Before any modelling, an examination of the empirical densities guides three structural choices. (i) *Symmetry.* `CO2_WLTP` is relatively symmetric and can be handled as is; `HORSEPOWER` has a heavy tail, but one attributable to residual outliers; `ENGINE_SIZE` concentrates around a few modes (standardised displacements), a sharp partition that argues for tree-based models. (ii) *Cross-market heterogeneity.* The distributions of `ENGINE_SIZE` and `CO2_WLTP` vary substantially across markets, as does the fuel mix (penetration of electric `EL` and hybrid `MX` vehicles): this is the main driver of the dataset shift analysed in §3. (iii) By contrast, the joint `HORSEPOWER`–`ENGINE_SIZE` distribution is similar across markets and displays clear clusters that learning can exploit.

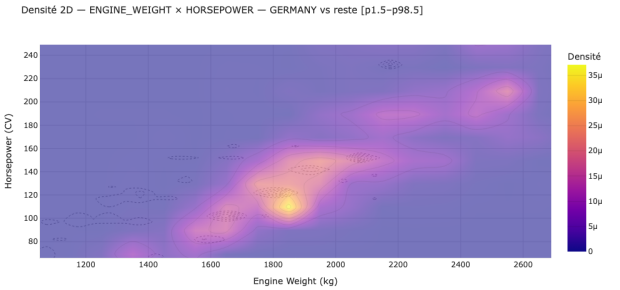


Figure 2. Exploratory analysis (German market): densities of the technical variables and fuel mix.

2.2 Models and explainability

We compare four models (Table 2): LightGBM [6] in two feature scopes (full and light scope), XGBoost [2], and an MLP. The detailed background (second-order boosting, the Adam optimiser, the algorithmic differences between LightGBM and XGBoost) is gathered in Appendix A.

Table 2. Characteristics of the four models compared.

Details	Model 1	Model 2	XGB	MLP
Algorithm	LightGBM	LightGBM	XGBoost	MLP
Features	full[a]	light[b]	light	light
`n_estim.`	3000	3000	3000	500
`lr`	0.05	0.05	0.05	10^{-3}
`max_depth`	6	4	6	–
`num_leaves`	31	15	–	–
`min_child`	20	50	5	–
Archi.	–	–	–	(128,64)
Early stop.	50	50	50	20

[a] `BRAND`, `MODEL`, `NEW_2HAND`, `FUEL_TYPE`, `ENGINE_WEIGHT`, `ENGINE_SIZE`, `HORSEPOWER`.
[b] `FUEL_TYPE`, `ENGINE_WEIGHT`, `ENGINE_SIZE`, `HORSEPOWER`.

2.2.1 SHAP and auditability

The requirements call for an **auditable methodology**: every prediction must be justifiable and verifiable by a third party (regulator, auditor, business team), failing which the model, however accurate, cannot be used in production. SHAP values [9, 16] provide exactly this auditability lever. Coming from cooperative game theory, they answer the question: if several "players" (the features) cooperate to produce a gain (the prediction), what is the fair contribution of each [10]? The contribution of feature j on observation x is

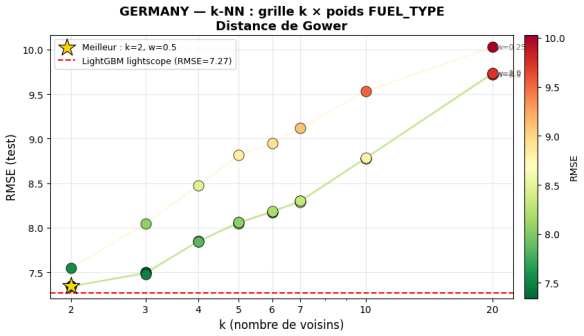
$$\phi_j(f,x) = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N|-|S|-1)!}{|N|!} \big[f_x(S \cup \{j\}) - f_x(S) \big], \tag{3}$$

with $f_x(S) = \mathbb{E}[f(X) \mid X_S = x_S]$: this is the average marginal gain brought by j, averaged over all subsets S of the other features. SHAP values are the unique attribution satisfying the axioms of efficiency, symmetry, linearity and null player, which sets them apart from heuristic importance measures. Their exact computation, exponential in general, becomes tractable on tree-based models thanks to the TreeSHAP algorithm, in $O(TLD^2)$ for T trees of depth D and L leaves, a decisive point when auditing a model over an entire portfolio. The beeswarm plot displays the distribution of the ϕ_j over the validation set, coloured by feature value: it reveals at a glance the global importance, the direction and the monotonicity of each effect (a high `HORSEPOWER` pushes the prediction upwards, and so on). Beyond *a posteriori* auditing, we use SHAP as an active *diagnostic* tool for model selection (§2.3): it is the inspection of the contributions that reveals which features harm the model.

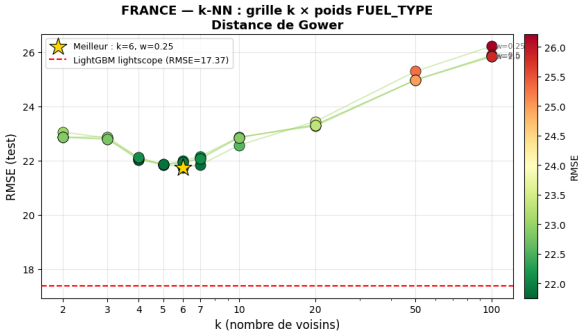
2.3 Comparative results

2.3.1 A first approach: k-NN regression

We first sought to exploit local similarity: two mechanically similar vehicles should emit similar amounts of CO_2, and a local non-parametric estimator could capture this relation without any parametric assumption on its form. The main difficulty is the definition of a distance on a mixed numerical-categorical space. To avoid an arbitrary one-hot encoding, we adopt the Gower distance, which normalises numerical features by their range and handles `FUEL_TYPE` through a $0/1$ distance, with a hyperparameter w controlling the relative weight of `FUEL_TYPE`. Selection is carried out by grid search over $(k,w) \in \{2,3,4,5,6,7,10,20\} \times \{0.25,0.5,1,2,4\}$, on a test set of 3000 observations (training limited to 4000 to contain the quadratic cost of the Gower matrix).



(a) German market.



(b) Cross-check market.

Figure 3. k-NN regression (Gower distance): RMSE as a function of (k,w). The behaviour differs radically from one market to the other, which discourages any generalisation.

On Germany (Fig. 3a), the RMSE increases monotonically with k, with no local minimum: the best configuration (k=2, w=0.5) reaches 7.34 g/km, against 7.27 g/km for light-scope LightGBM. This performance, good in absolute terms, says nothing about generalisation ability, and such a low k is not robust. The cross-check on a second market (Fig. 3b) reveals a radically different behaviour: the RMSE decreases and then increases, with a weakly marked interior minimum around k=6, and w has almost no effect (a sign that `FUEL_TYPE` is already separated in the numerical space); the best RMSE there is 21.8 g/km, that is 4.38 g/km above LightGBM. This result is paradoxical because the neighbourhoods are genuinely local: the ratio $\bar{d}_{k_{\text{opt}}}/\text{diam}(\mathscr{X})$ equals 0.03 and 0.01 on the two markets, meaning that the k_{opt} nearest neighbours cover less than 3 % and 1 % of the diameter of the space. The explanation lies in the *discrete* structure of the data: the space is not a regular continuum but a set of clusters separated by sharp boundaries (electric vs. combustion, large displacements vs. the rest of the fleet). In these transition zones, two neighbours in the Gower sense can have very different intensities, which makes the average of the neighbours uninformative, precisely where a decision tree excels by turning those boundaries into priority splits. Locality is therefore necessary but not sufficient: the target must also be *regular* within the neighbourhood, which is not the case here. The differential variant DNNR [12], which corrects discontinuities through a local Taylor expansion, does not apply to this topology. k-NN is therefore ruled out: poorly

auditable (interpolating an emission from two neighbours is unacceptable in a banking context), unstable across markets, and computationally costly, since it requires a linear scan over a dataset this large.

2.3.2 A paradigm shift: tree-based models

The sharp boundaries in the data call for a decision-tree architecture. We proceeded by trial and error, from a general-purpose model towards more parsimonious variants, then an MLP as a control. The results concern Germany (the cleanest database; `BELGIUM` and `NETHERLANDS` are unusable, as features are missing there). Four criteria guide the selection: **accuracy** (cross-validated RMSE), **overfitting risk** (train/validation gap), **geometric fidelity** (CO_2 heatmaps in a feature plane, which must neither overload nor underload any region and must recover the zero emissions of electric vehicles) and **explainability** (SHAP). The complete quantitative results are in Table 3; the consistency heatmaps of each model are reported in Appendix B.

Table 3. Performance of the four models on Germany (WLTP and NEDC targets). h.-o.: hold-out.

Quantity	M1	M2	XGB	MLP
Target: CO_2 WLTP				
Train RMSE (h.-o.)	3.39	6.37	4.22	11.93
Val RMSE (h.-o.)	5.21	7.32	5.90	11.94
Gap (val−train)	1.82	0.95	1.68	0.01
CV RMSE (±std)	5.56 ±.32	7.27 ±.23	5.73 ±.23	13.06 ±.44
CV MAE	2.94	4.36	2.94	∼9.5
CV R^2	.9911	.9860	.9913	∼.92
95% CI (±g/km)	11	14	11	26
Target: CO_2 NEDC				
Train RMSE (h.-o.)	7.37	9.57	6.17	14.76
Val RMSE (h.-o.)	12.74	11.02	9.73	14.50
Gap (val−train)	5.37	1.45	3.56	−0.26
CV RMSE (±std)	13.73 ±1.3	10.99 ±.14	9.68 ±.23	15.29 ±.29
CV MAE	8.71	7.44	5.80	11.22
CV R^2	.8996	.9349	.9495	.8742
95% CI (±g/km)	27	22	19	30

Model 1 (full-scope). The model shows significant overfitting, exacerbated under NEDC (validation−train gap of 1.82 under WLTP, 5.37 under NEDC). The SHAP values (Fig. 4) reveal its origin: the contributions of `NEW_2HAND`, `BRAND` and `MODEL` are tightly packed and show no directional pattern, whereas the mechanical variables follow a logical pattern (a low displacement, weight or power lowers the prediction, a high value raises it; `FUEL_TYPE`, being categorical, forms clean, compact and disjoint clusters). These three features are in fact dangerous on more than one count. First, they are at best useless: the intensity in the databases is a factory-gate value, so `NEW_2HAND` (new/second-hand) should not modify it. Second, they are dangerously redundant: brand and model act on emissions only through the physical characteristics (displacement, weight, power) that are already present. Finally, `BRAND` and `MODEL` have high cardinality (hundreds of labels): at each node the tree evaluates hundreds of candidate splits, which increases the probability of finding a good split *by chance* on the training sample, even in the absence of any real predictive power. Contrary to the intuition that "more variables means better prediction", this increases overfitting if those variables are nothing but noise.

Model 2 (light-scope). In the light of this diagnosis, we remove the variables of no interest. The performance is markedly better: overfitting is far lower, the error is barely higher under WLTP but lower under NEDC, and remains very satisfactory throughout. The SHAP values (Fig. 5) are reassuring, with a clear directional pattern and strongly marked clusters (an electric cluster on `FUEL_TYPE` and `ENGINE_SIZE`). The heatmaps (Appendix B) confirm that the initial geometry is respected: the regions of very low intensity (electric vehicles, zero displacement), of very high intensity (large displacements, characteristic of the German market) and the intermediate regions are faithfully recovered. These heatmaps have a heuristic value: they locate the poorly predicted zones without constituting absolute proof. Model 2 achieves the best bias-variance trade-off and meets the auditability requirement.

XGBoost and MLP. The two remaining models confirm this choice. XGBoost (Fig. 6) reaches a lower final RMSE but reintroduces non-negligible, even pronounced, overfitting; its SHAP values show that `HORSEPOWER` loses predictive power under NEDC while `FUEL_TYPE` gains some, the directional pattern being broadly preserved. In general, LightGBM outperforms XGBoost thanks to its leaf-wise growth: it identifies and deepens the most informative leaves first (the discontinuity of electric vehicles, $CO_2 \approx 0$), whereas the level-wise growth of XGBoost dilutes the split budget over regions that are already well explained. The MLP, despite intensive feature tuning, systematically underfits: its accuracy is heavily degraded. This confirms the known superiority of trees on tabular data [17]. [4] gives two explanations for it. On the one hand, MLPs are biased towards smooth target functions (spectral bias towards low frequencies [15]): Adam first reduces the global error, and learns local variations exponentially more slowly, yet our targets vary violently (a 300 hp hybrid can emit 30 g/km, a combustion vehicle of the same power and displacement up to 200 g/km). On the other hand, MLPs are rotationally invariant: each neuron linearly combines all the features, so that an uninformative feature perturbs the learning of the true predictors; [14] shows that the sample complexity of any rotationally invariant algorithm grows linearly with the number of useless features, whereas trees, which evaluate each feature separately, are structurally immune to this. **In the end, Model 2 (LightGBM light-scope) is retained.**

2.3.3 Cross-market comparison

We extend the comparison to the usable markets (Table 4). *Consistency of the metrics*: the between-fold standard deviations are low (below 6 % of the mean RMSE), a sign of good stability; the relatively high RMSE/MAE ratio signals a few predictions with large errors, most likely atypical configurations (plug-in hybrids) that the four light-scope features do not capture perfectly. *Cross-model comparison*: XGBoost produces the lowest RMSE in Spain (WLTP and NEDC), at times with a marked advantage, but always at the cost of non-negligible overfitting; Model 2 achieves the best bias-

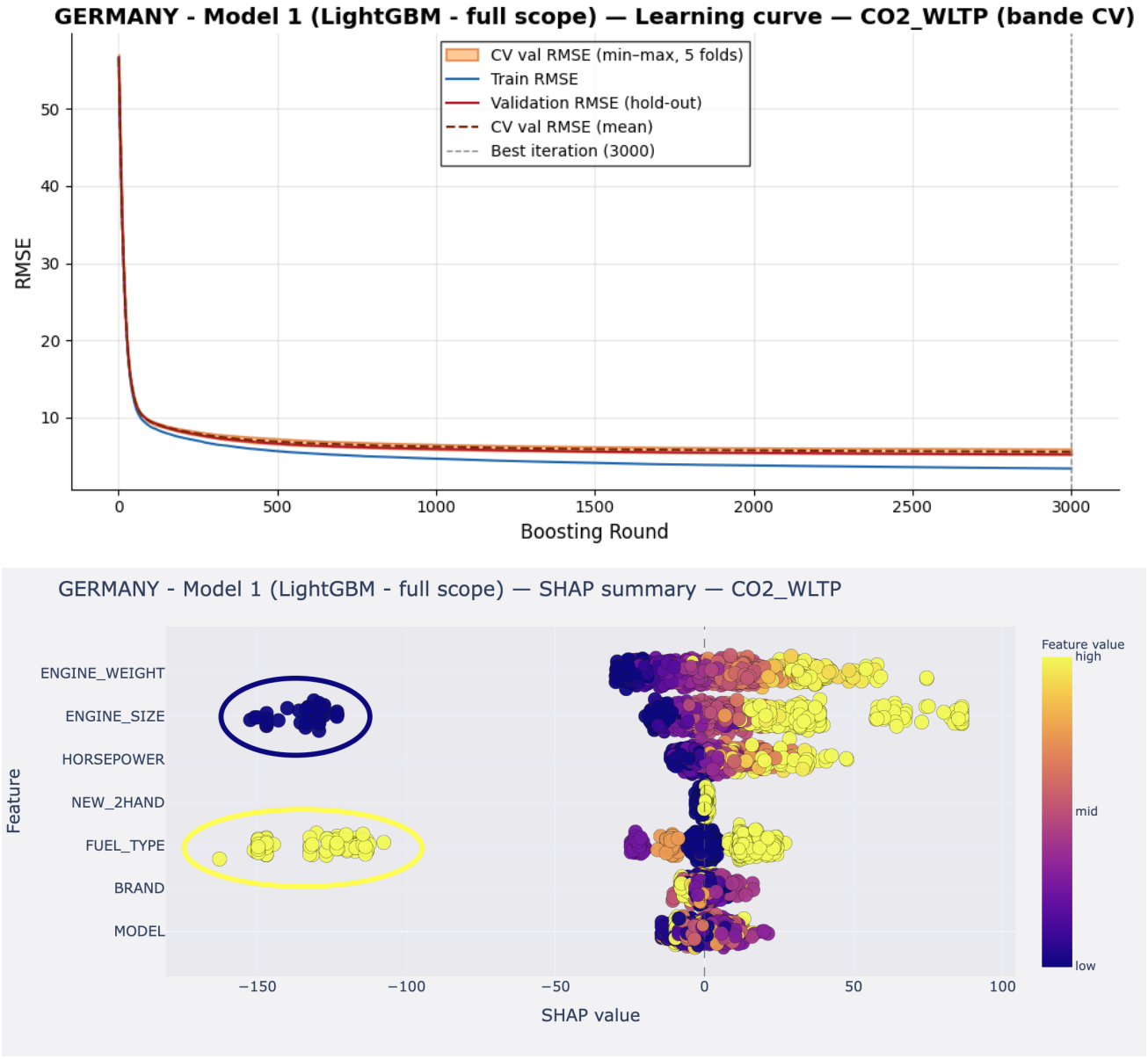
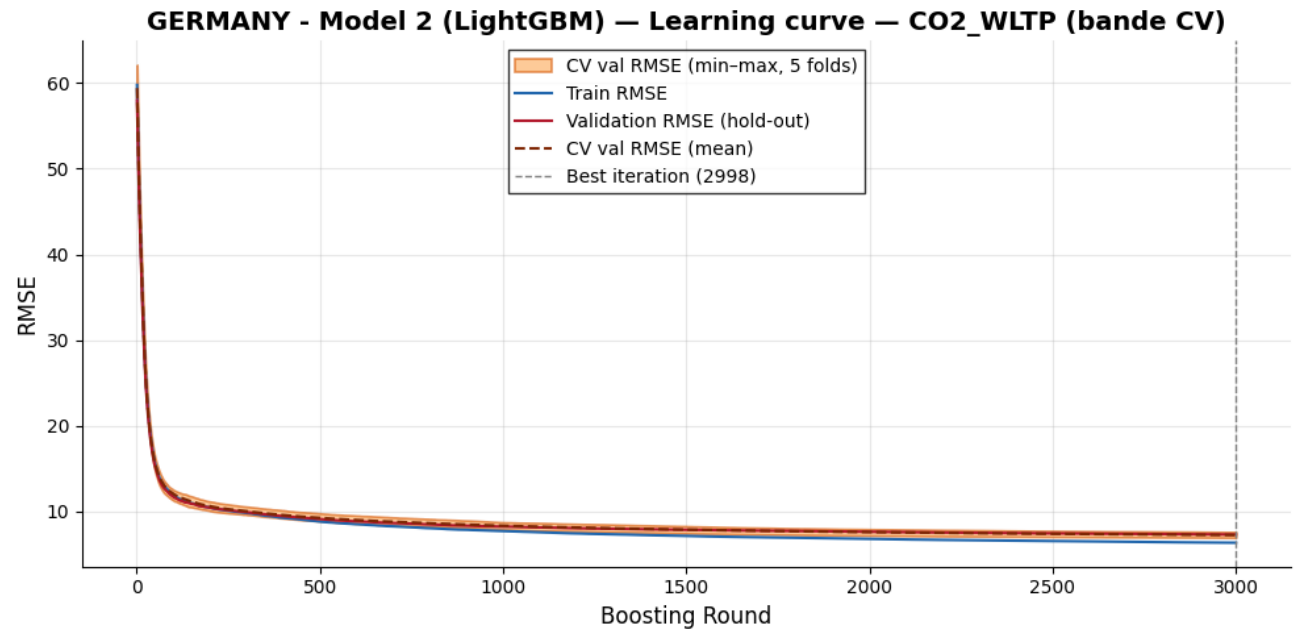


Figure 4. Model 1 (full-scope), Germany, WLTP target. Top: learning curve; bottom: SHAP beeswarm. Overfitting is visible; `BRAND`, `MODEL` and `NEW_2HAND` show packed, non-directional SHAP contributions.



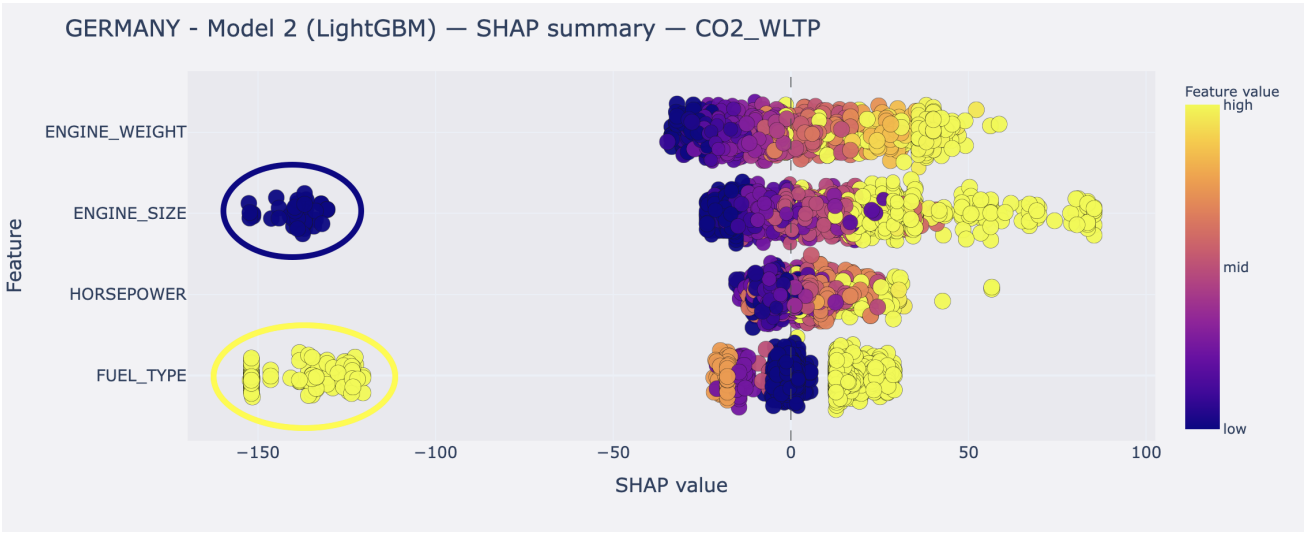
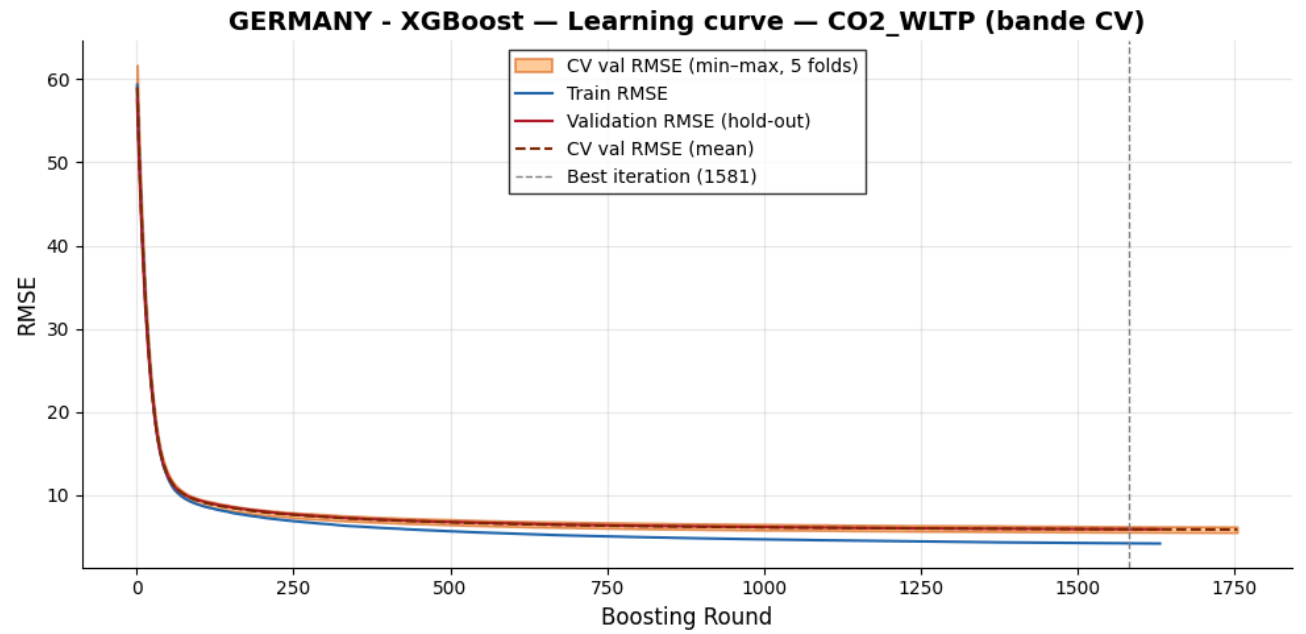


Figure 5. Model 2 (light-scope), Germany, WLTP target. Top: learning curve; bottom: SHAP beeswarm. Overfitting is contained, and the contributions are directional and readable (a clean electric cluster on `FUEL_TYPE` and `ENGINE_SIZE`).



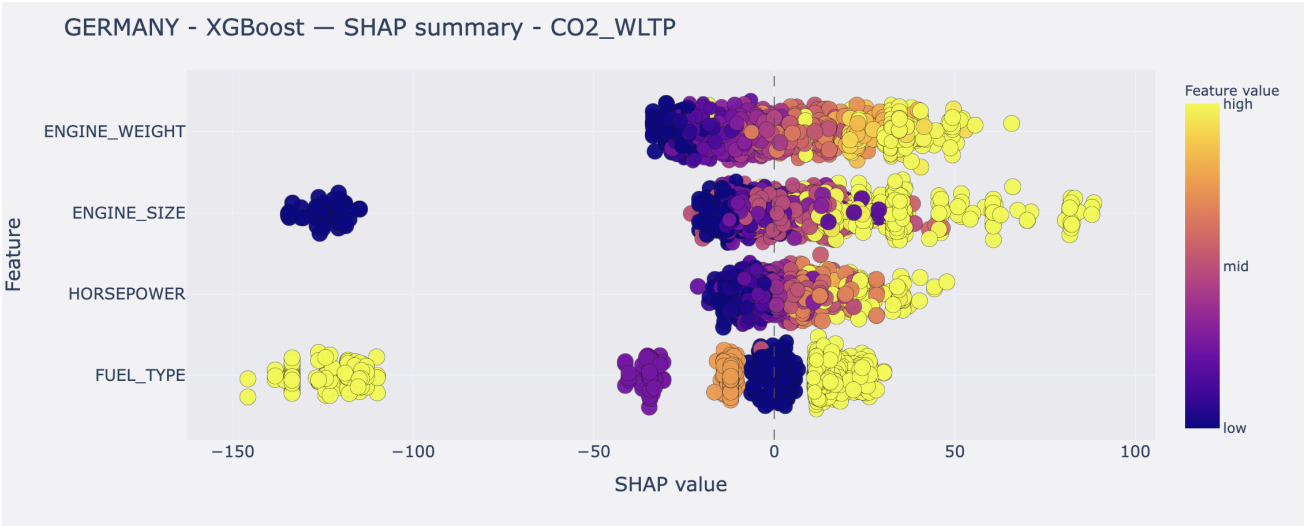


Figure 6. XGBoost (light-scope), Germany, WLTP target. Top: learning curve; bottom: SHAP beeswarm. Lower final RMSE, but overfitting is reintroduced.

variance trade-off and dominates on Germany. The MLP is systematically the weakest, with an RMSE 1.3 to 1.8× that of Model 2; the only near-exception is Spain under WLTP (MLP 9.32 vs. Model 2 9.42), where the large sample size (159 142 obs.) partially compensates the spectral bias, the MLP/tree gap narrowing with dataset size, as [4] predicted. *Cross-market comparison*: contrary to intuition, dataset size does not significantly reduce the error (trees work as soon as #obs ≫ #features, which always holds here); on the other hand the fleets differ in homogeneity, some of them showing more variance and isolated, poorly predicted segments. Above all, **the same model (Model 2) prevails whatever the market**, which is operationally reassuring: the structural differences do not justify training distinct types of model per country. *WLTP vs. NEDC*: the NEDC standard, older and less technologically discriminating, produces more regular values, which are therefore often easier to predict from the mechanical features; the effect reverses, however, wherever the number of NEDC observations is small.

Table 4. Cross-validation (5 folds) by market and by standard: RMSE ± between-fold standard deviation, MAE, and bias-variance diagnosis.

	Metric	Model 2	XGBoost	MLP
Germany	$\text{RMSE}_{\text{WLTP}}$	**7.27**$_{\pm.23}$	5.73$_{\pm.23}$	13.06$_{\pm.44}$
	MAE_{WLTP}	**4.36**	2.94	8.88
	$\text{RMSE}_{\text{NEDC}}$	**10.99**$_{\pm.14}$	9.68$_{\pm.23}$	15.29$_{\pm.29}$
	MAE_{NEDC}	**7.44**	5.80	11.22
	Diag.	**G**	MO/SO	UF
Spain	$\text{RMSE}_{\text{WLTP}}$	9.42$_{\pm.16}$	**6.36**$_{\pm.12}$	9.32$_{\pm.19}$
	MAE_{WLTP}	4.10	**3.73**	5.87
	$\text{RMSE}_{\text{NEDC}}$	7.36$_{\pm.39}$	**6.57**$_{\pm.29}$	9.58$_{\pm.57}$
	MAE_{NEDC}	3.11	**2.46**	5.10
	Diag.	**G**	G/MO	UF

G: good trade-off; MO/SO: mild / severe overfitting; UF: underfitting.

2.3.4 Conformal intervals (MAPIE)

Beyond scalar metrics, auditability requires an uncertainty bound. Conformal intervals (Appendix A) are directly actionable: Model 2 produces 95 % intervals of ±14 g/km (Germany WLTP) to ±22 g/km (Germany NEDC). In concrete terms, a prediction of 142 g/km can be accompanied by an upper bound guaranteed at 95 %, usable in a conservative carbon exposure computation, in line with CSRD. This coverage guarantee is distribution-free, making no assumption on the shape of the residuals, which makes it robust to heteroscedasticity of the error across fleet segments.

Avenues for improvement. Several directions would increase accuracy or usability: enriching the features with external homologation data (manufacturer catalogues, ADEME) after checking their predictive power through SHAP; Bayesian hyperparameter optimisation (Optuna, Tree-structured Parzen Estimator), more efficient than a grid search; and stacking of models whose errors are structurally decorrelated (leaf-wise LightGBM, level-wise XGBoost, MLP), with a linear meta-model weighting their out-of-fold predictions so as to avoid overfitting the meta-model.

3 Towards an international model?

Motivations. The partner operates in many European markets. An aggregated international model would offer three structural advantages over independent national models: (i) a far larger training set, reducing the variance of the estimates for rare or recent vehicles; (ii) redundant observations for vehicle models common to several markets (shared European platforms), which strengthens robustness; (iii) simpler governance and maintenance. But this aggregation is legitimate only if the covariate distributions $P_c(X)$ are sufficiently close across markets: a model trained on one distribution and applied to another does not generalise reliably. We therefore test this feasibility before making any recommendation.

Domain classifier. Following [8], we train a multiclass LightGBM to predict the *market of origin* of a vehicle from its technical characteristics alone (`ENGINE_SIZE`, `ENGINE_WEIGHT`, `FUEL_TYPE`, `HORSEPOWER`, `CO2_WLTP`). Including the target is not problematic here: we want to know not only whether the vehicles differ, but also whether the vehicle–emissions *relation* differs across markets. The learning curve (Fig. 7) shows no significant overfitting, and the accuracy reaches ≈ 96 %: to validate an international model, an accuracy close to chance ($1/K$ for K markets) would have been required.

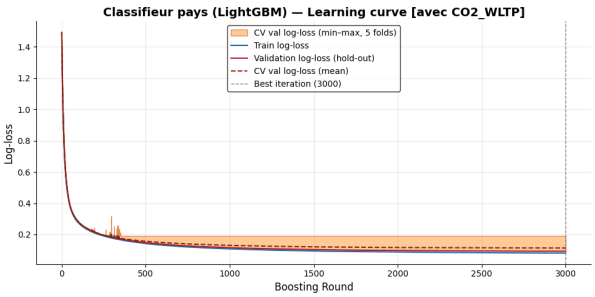


Figure 7. Domain classifier (multiclass LightGBM): learning curve. No significant overfitting.

The confusion matrix (Fig. 8) confirms that the technical characteristics make it possible to identify the market of origin with high reliability, some markets standing out very strongly. This result may come as a surprise, since *a priori* nothing distinguishes the overall car market of one country from another, but the sample reflects commercial and strategic lending policies specific to each market.

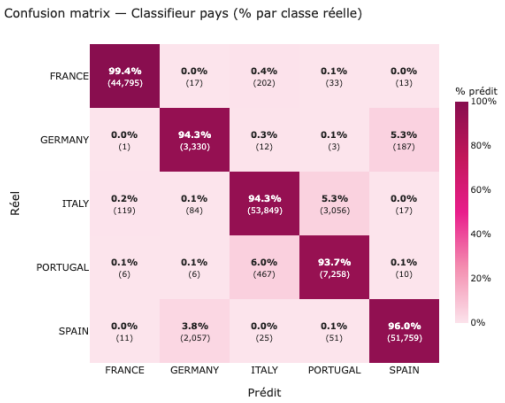


Figure 8. Confusion matrix of the domain classifier. The markets are identified with high reliability.

PCA and clustering of the markets. To understand this discriminability, we project the data by principal component analysis (Fig. 9). On the first three components (which explain $\approx 60\,\%$ of the variance), distinct clusters by market appear clearly. The fit is performed on a stratified sample of about 50 000 rows per market, and we display the cluster centroids obtained by k-means; mean-shift smoothing was tested to attenuate the noise but degraded the spatial separation. Elongated and spatially isolated clusters are observed in particular for certain markets. This geometric structure directly corroborates the performance of the classifier: the markets occupy distinct regions of the feature space.

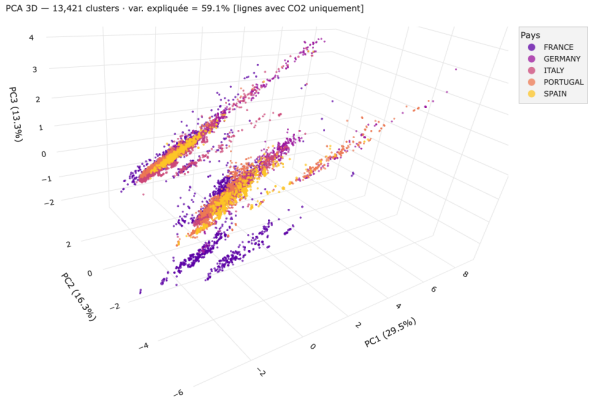


Figure 9. PCA projection (three components, $\approx 60\,\%$ of the variance) and k-means centroids by market: distinct clusters appear.

Covariate shift detection. To quantify precisely, for each market c, whether $P_c(X) \neq P_{\text{global}}(X)$, we train a binary LightGBM classifier separating the observations of c (label 0) from those of the global dataset (label 1). The AUC measures the separability of the two distributions (close to 0.5: no detectable shift). We test two variants: on the features alone (X), and on features + target (X, Y), the latter detecting a possible concept shift (a variation of $P(Y \mid X)$ across markets), which is more serious than a mere shift in the characteristics. The results (Table 5) show a **marked covariate shift** (AUC from 0.82 to 0.98 depending on the market) but $\text{AUC}(X) \approx \text{AUC}(X+Y)$: adding the target does not increase separability, hence **no concept shift**. The emissions–features relation is stable across markets; only the feature distributions differ.

Table 5. AUC of the binary domain classifier by market (X: features alone; $X+Y$: with the target `CO2_WLTP`).

Country	AUC (X)	AUC ($X+Y$)	Interpretation
Germany	0.9841	0.9863	very strong shift
Portugal	0.9656	0.9737	very strong shift
France	0.8586	0.8579	strong shift
Spain	0.8335	0.8327	strong shift
Italy	0.8201	0.8251	strong shift

Correction by reweighting and feasibility. Since the problem reduces to a pure covariate shift, it is correctable by importance reweighting [18]: each observation of market c receives a weight $w(x) = \hat{p}(x)/(1-\hat{p}(x))$, where $\hat{p}(x)$ is the probability of belonging to the global dataset given by the binary classifier (interpretation: likelihood ratio p_{global}/p_c). The weights are clipped at the 99th percentile and normalised to mean 1, then passed through `sample_weight`, bringing the training distribution closer to the deployment distribution. Our operational recommendations follow from this: an international model is directly applicable for markets with a low AUC (< 0.70), with a net gain in training size; a national model, or a reweighted international one, for markets with a strong shift (> 0.80). One point of caution: three markets concentrate $\approx 75\,\%$ of the data, so that the global model is in effect the model of those markets, and performance on an under-represented market must be interpreted with care, whatever the AUC.

4 Conclusion

This work establishes a complete and *auditable* methodology for predicting the carbon intensity of a real automotive portfolio that is multi-market and heterogeneous, a setting far removed from the clean homologation databases of the literature and therefore carrying difficulties of its own.

Theoretical conclusions. In this tabular regime, boosted trees dominate: LightGBM on a restricted feature scope achieves the best bias-variance trade-off and systematically outperforms an MLP, in line with the literature and for structural reasons (the spectral bias and rotational invariance of dense networks, and the leaf-wise growth of LightGBM being better suited to discontinuities in the data). Conformal prediction (MAPIE) provides statistically valid intervals under the sole assumption of exchangeability, without recourse to normality or homoscedasticity of the residuals. Finally, the dataset shift analysis establishes a clear result: a strong cross-market covariate shift but no concept shift, which makes the transfer correctable by reweighting without relearning the underlying relation.

Practical conclusions. For deployment, Model 2 (LightGBM light-scope) is the retained candidate: accurate, robust, and above all auditable, the same model prevailing on every market. SHAP and the conformal intervals provide the two auditability building blocks required by the regulatory framework (CSRD). The international model is not directly transposable as it stands, but a hybrid strategy (national /

reweighted international depending on the shift AUC) is actionable.

Difficulties encountered and methodological path. The approach was not linear. The degraded quality and the heterogeneity of the data called for pre-processing and controlled imputation upstream. Model selection proceeded by trial and error: a first local approach (k-NN) failed because of the discrete, sharp-boundary structure of the data; a first general-purpose tree model suffered from overfitting, and it was the SHAP inspection, used as a *diagnostic* tool rather than as mere *a posteriori* explanation, that revealed the harm done by redundant and high-cardinality features, motivating the decisive scope reduction. The need for auditability, present throughout, guided every trade-off between raw accuracy and usability. The main extension remains the prediction of the annual distance travelled, which is absent from the data and needed to compute the total footprint. By placing auditability and uncertainty quantification on the same footing as accuracy, the methodology aims to remain relevant well beyond the application that motivated it.

A Theoretical complements

We assume the reader is familiar with the fundamentals of tree-based models, gradient boosting, the MLP and cross-validation, and we restrict this appendix to the points specific to our choices.

Gradient boosting. We fit $f(x) = \sum_{t=1}^{T} \eta\, f_t(x)$, a weighted sum of CART trees, each f_t correcting the residuals of the preceding sum through a second-order Taylor expansion of the loss: with g_i, h_i the gradient and Hessian, the optimal weight of a leaf and the gain of a split are written $w_j^* = -G_j/(H_j+\lambda)$ and $\text{Gain} = \frac{1}{2}[G_L^2/(H_L+\lambda) + G_R^2/(H_R+\lambda) - (G_L+G_R)^2/(H_L+H_R+\lambda)] - \gamma$, where G_j, H_j sum g_i, h_i over the leaf.

LightGBM vs. XGBoost. Both share this framework, but differ on four decisive points. *(1) Histogram-based splitting*: LightGBM [6] discretises each continuous feature into $k \leq 255$ bins, bringing the cost of the split search down from $O(n \log n)$ to $O(k)$ per feature; a histogram subtraction trick ($\mathscr{H}_R = \mathscr{H}_{\text{parent}} - \mathscr{H}_L$) halves this cost again. *(2) GOSS* (Gradient-based One-Side Sampling): the $a\%$ of observations with large $|g_i|$ (the most informative ones) are kept in full and a fraction b of the small-gradient observations is subsampled, their gradients being multiplied by $\frac{1-a}{b}$ to keep the gain estimator unbiased, so that training is carried out on 20–40 % of the data at almost constant accuracy. *(3) EFB* (Exclusive Feature Bundling): mutually exclusive features (typically arising from a one-hot encoding) are merged by graph colouring into a single bundle, reducing the effective dimension from p to $p' \ll p$. *(4) Leaf-wise growth*: this is the most structural difference. XGBoost [2] grows level-wise (all nodes of a level are expanded before the next one, giving symmetric trees), whereas LightGBM grows leaf-wise (at each iteration the leaf of maximum gain is split, whatever its depth). For a given split budget, LightGBM reduces the loss faster and concentrates its capacity on the informative regions, for instance the discontinuity of electric vehicles ($CO_2 \approx 0$), which is identified and deepened immediately, whereas XGBoost dilutes part of the budget over low-residual regions. The overfitting risk of the leaf-wise strategy is controlled by `num_leaves` and `max_depth`.

The Adam optimiser. For the MLP, Adam [7] adapts the step size per weight through the first- and second-order moments of the gradient:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2, \tag{4}$$

$$w \leftarrow w - \frac{\eta}{\sqrt{\hat{v}_t} + \varepsilon}\, \hat{m}_t, \quad \hat{m}_t = \frac{m_t}{1-\beta_1^t},\ \hat{v}_t = \frac{v_t}{1-\beta_2^t}. \tag{5}$$

It is more robust to noise and faster than SGD, without prior tuning of the features.

Conformal intervals (MAPIE). Conformal prediction [21, 19, 3] assumes only *exchangeability* (weaker than i.i.d.), and requires neither normality nor homoscedasticity of the residuals. On a calibration set $\mathscr{D}_{\text{cal}}$ disjoint from the training set, we compute the non-conformity scores $s_i = |y_i - \hat{f}(x_i)|$, then the empirical quantile $\hat{q}_{1-\alpha}$; the interval for a new x is $\hat{C}_\alpha(x) = [\hat{f}(x) \pm \hat{q}_{1-\alpha}]$. In order not to sacrifice a fixed set and to reduce the variance, we embed MAPIE within each fold of the 5-CV (80 % fitting / 20 % calibration, `cv="prefit"`), for $1-\alpha \in \{80, 90, 95\}\%$, then aggregate. Interpretation: a 95 % interval of ± 14 g/km means that at least 95 vehicles out of 100, drawn from the same population, have their true CO_2 within the predicted interval.

B Consistency heatmaps (Germany, WLTP)

The heatmaps below represent CO_2 intensity in a feature plane (e.g. `ENGINE_WEIGHT`×`ENGINE_SIZE`). For each model, three planes are shown; each panel places the distribution *before* and *after* prediction side by side, with the associated colour scale. They serve as a heuristic check: a faithful model must neither overload nor underload any region, and must recover the zero emissions of electric vehicles.

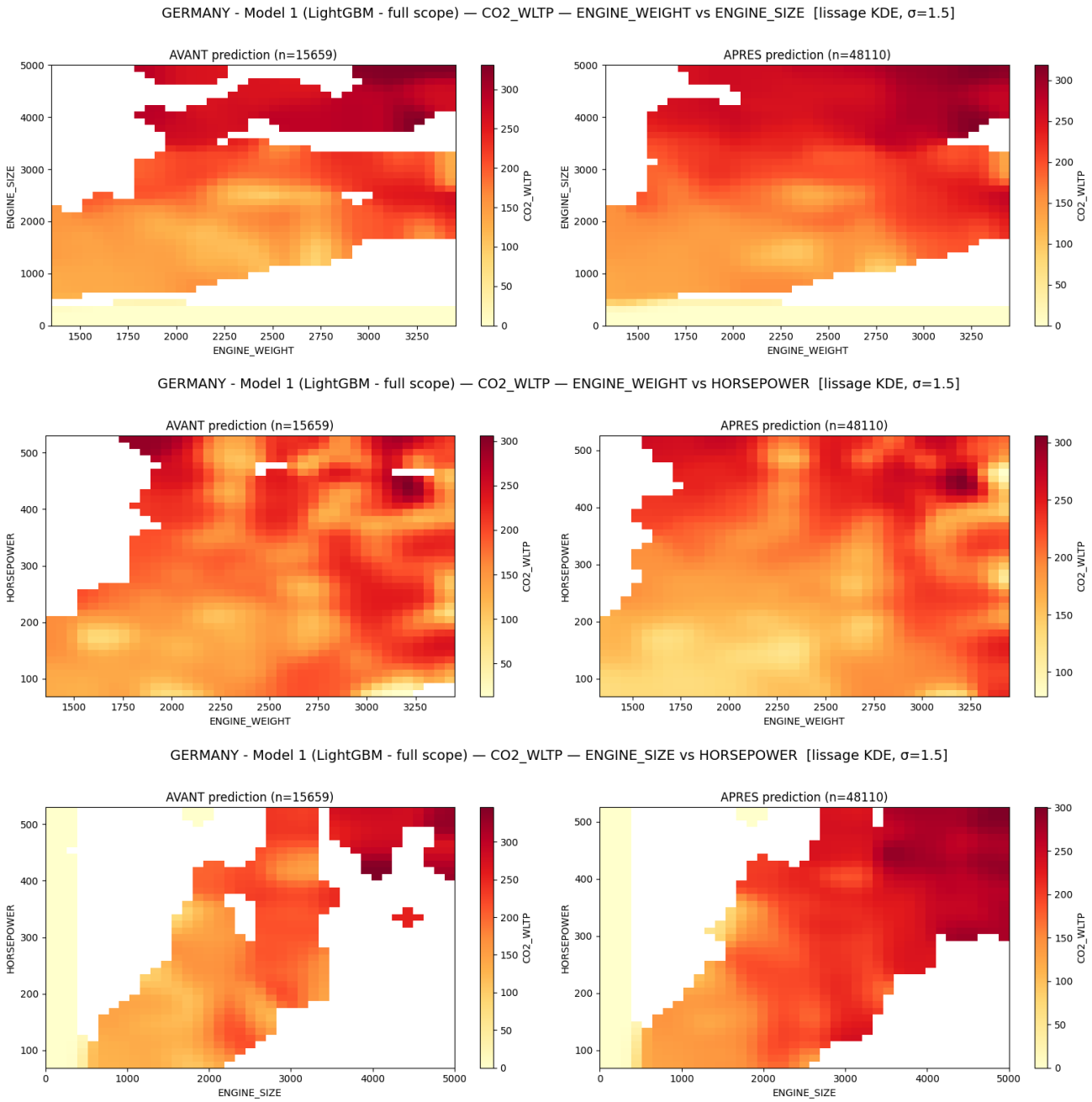


Figure 10. Model 1 (full-scope), Germany, WLTP target: CO_2 consistency heatmaps on three feature planes. Each panel compares the distribution *before* (left) and *after* (right) prediction.

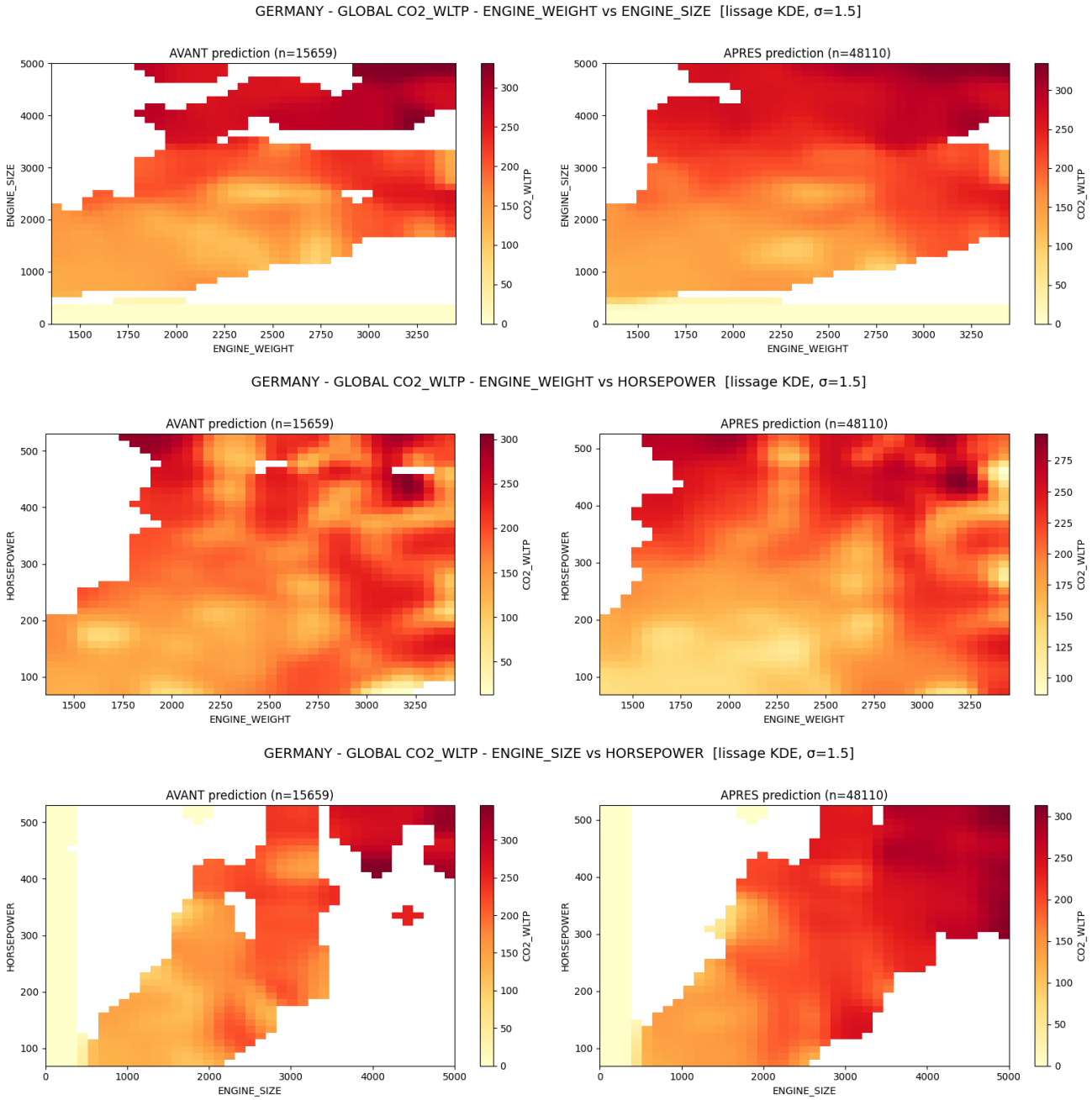


Figure 11. Model 2 (light-scope), Germany, WLTP target: CO_2 consistency heatmaps on three feature planes (*before* vs. *after* prediction). The initial geometry is faithfully recovered.

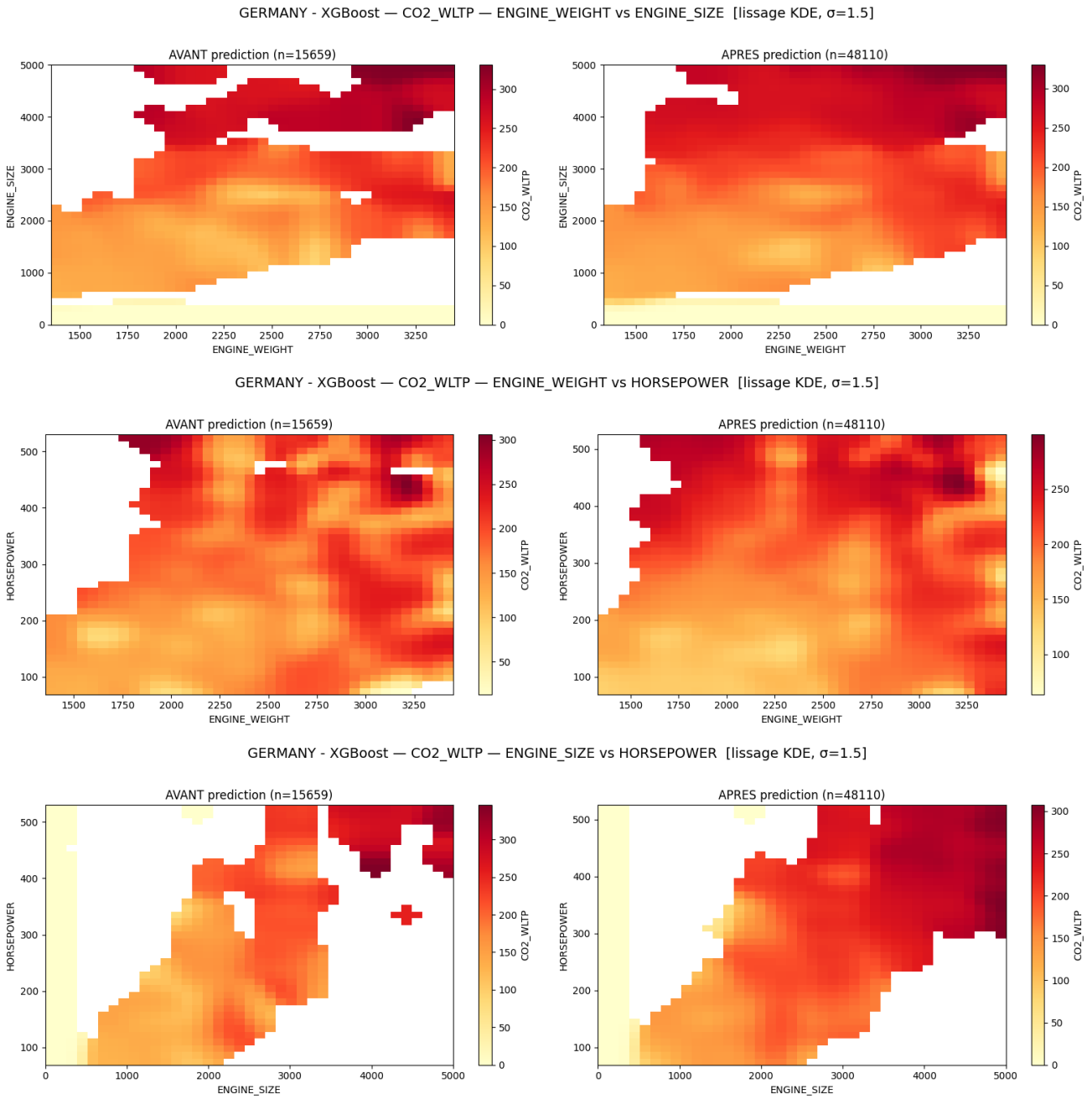


Figure 12. XGBoost (light-scope), Germany, WLTP target: CO_2 consistency heatmaps on three feature planes (*before* vs. *after* prediction).

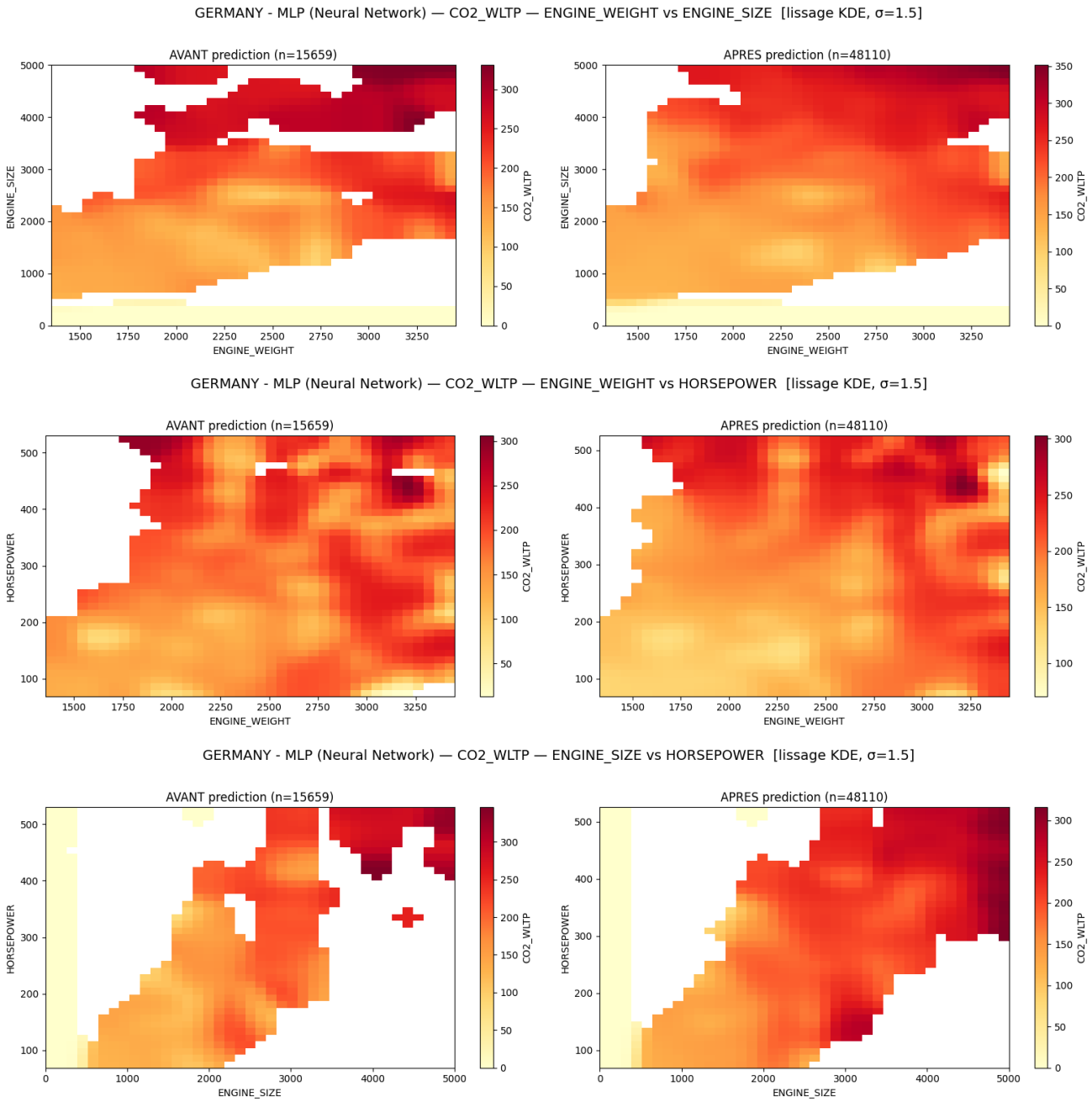


Figure 13. MLP, Germany, WLTP target: CO_2 consistency heatmaps on three feature planes (*before* vs. *after* prediction). The underfitting shows up as smoothed regions.

References

[1] Akinosho, T. D., Oyedele, L. O., & Davison, B. (2024). *Application of Machine Learning to Predict CO_2 Emissions in Light-Duty Vehicles*. Sensors, 24(24), 8219.

[2] Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*. Proc. 22nd ACM SIGKDD, 785–794.

[3] Cordier, T., Blot, V., Lacombe, L., Morzadec, T., Capitaine, A., & Brunel, N. (2023). *Flexible and Systematic Uncertainty Estimation with Conformal Prediction via the MAPIE library*. COPA, PMLR.

[4] Grinsztajn, L., Oyallon, E., & Varoquaux, G. (2022). *Why do tree-based models still outperform deep learning on typical tabular data?* NeurIPS 35, 507–520.

[5] Gurcan, F. (2024). *Forecasting CO_2 emissions of fuel vehicles using ensemble learning, machine learning, and deep learning models*. PeerJ Computer Science, 10, e2234.

[6] Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., Ye, Q., & Liu, T.-Y. (2017). *LightGBM: A Highly Efficient Gradient Boosting Decision Tree*. NeurIPS 30.

[7] Kingma, D. P., & Ba, J. (2015). *Adam: A Method for Stochastic Optimization*. ICLR 2015. arXiv:1412.6980.

[8] Lopez-Paz, D., & Oquab, M. (2017). *Revisiting Classifier Two-Sample Tests*. ICLR 2017. arXiv:1610.06545.

[9] Lundberg, S. M., & Lee, S.-I. (2017). *A Unified Approach to Interpreting Model Predictions*. NeurIPS 30.

[10] Lundberg, S. M., Erion, G., Chen, H., et al. (2020). *From local explanations to global understanding with explainable AI for trees*. Nature Machine Intelligence, 2, 56–67.

[11] Muraru, M. M., Simó, Z., & Iantovics, L. B. (2025). *Study on State-of-the-Art ML Algorithms Efficiency for CO_2 Emissions Prediction in Fleet Management Optimization*. LNNS, vol. 1249, Springer.

[12] Nader, Y., Sixt, L., & Landgraf, T. (2022). *DNNR: Differential Nearest Neighbors Regression*. arXiv:2205.08434.

[13] Natarajan, Y., Wadhwa, G., Sri Preethaa, K. R., & Paul, A. (2023). *Forecasting Carbon Dioxide Emissions of Light-Duty Vehicles with Different Machine Learning Algorithms*. Electronics, 12(10), 2288.

[14] Ng, A. Y. (2004). *Feature selection, L1 vs. L2 regularization, and rotational invariance*. ICML 2004, art. 78.

[15] Rahaman, N., Baratin, A., Arpit, D., et al. (2019). *On the Spectral Bias of Neural Networks*. ICML 2019, PMLR 97, 5301–5310.

[16] Shapley, L. S. (1953). *A Value for n-Person Games*. Contributions to the Theory of Games, vol. 2, 307–317. Princeton Univ. Press.

[17] Shwartz-Ziv, R., & Armon, A. (2022). *Tabular Data: Deep Learning is Not All You Need*. Information Fusion, 81, 84–90.

[18] Sugiyama, M., Krauledat, M., & Müller, K.-R. (2007). *Covariate Shift Adaptation by Importance Weighted Cross Validation*. JMLR, 8(35), 985–1005.

[19] Taquet, V., Blot, V., Morzadec, T., Lacombe, L., & Brunel, N. (2022). *MAPIE: an open-source library for distribution-free uncertainty quantification*. arXiv:2207.12274.

[20] Tena-Gago, D., Serna, A., & Lozano-Diez, A. (2023). *Machine-Learning-Based Carbon Dioxide Concentration Prediction for Hybrid Vehicles*. Sensors, 23(3), 1563.

[21] Vovk, V., Gammerman, A., & Shafer, G. (2005). *Algorithmic Learning in a Random World*. Springer.